Abdelrahman Khalifa

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Summary

Embedded software engineer with over a year of professional C firmware work on 32-bit microcontrollers: owns the Zephyr RTOS firmware for EKSON's pneumatic motion simulators. Automotive background from an AUTOSAR-based V2X graduation project on STM32 (UDS firmware updates and a bootloader with CRC check), AUTOSAR MCAL drivers and a FreeRTOS car seat-heater controller. Merged contributions to Zephyr RTOS and The Linux Foundation's OpenPrinting. Works in C, C++ and Python; based in Cairo.

Experience

Embedded Software Engineer Aug 2025 – Present
EKSON *Egypt*

- Wrote the ESP32-S3 firmware and migrated it from ESP-IDF/FreeRTOS to Zephyr RTOS.
- Implemented closed-loop PID control of the pneumatic valves, with feedback from an MPU6050 IMU over I2C, plus safety stops and a watchdog.

Google Summer of Code Contributor May 2026 – Oct 2026
The Linux Foundation (OpenPrinting) *Remote*

- Built the **Printers page for COSMIC Settings** as 5 Rust crates, with the C backend isolated behind a varlink IPC service (cosmic-settings #2182, in review).
- **11 PRs merged upstream** (cups-rs, #164, #165).

Embedded Linux Systems Intern Dec 2024 – Jan 2025
Orange Digital Center (ODC) *Egypt*

- Built Linux images with Yocto and Buildroot and a kernel module; scored 100% in the final evaluation.

OSS Projects

Zephyr RTOS Mar 2025

- Implemented the `IP_MULTICAST_LOOP` socket option for IPv4 across 6 networking modules, reviewed and merged upstream (issue #86977).

Projects

AUTOSAR-Based V2X Vehicle | *graduation project, team; STM32, ESP32* GitHub

- Owned the firmware update path. An ESP32 gateway downloads the image and sends it over UART with UDS services (session control, request download 0x34, transfer data 0x36, transfer exit 0x37, read version); the STM32 bootloader verifies it with the hardware CRC unit, keeps the version in flash and uses two banks with rollback.

AUTOSAR PORT and GPIO drivers | *C, TM4C*

- MCAL drivers developed and tested against the AUTOSAR specification.

Seat Heater Control System | *Tiva C, FreeRTOS* GitHub

- Heater controller for two car seats with event groups, a mutex-protected UART display, sensor-failure diagnostics and run-time measurements (CPU load, task times).

Technical Skills

Languages: C, Modern C++, Python, Rust, Bash
Automotive & RTOS: AUTOSAR (MCAL), UDS diagnostics and flashing, bootloaders, FOTA, OSEK/VDX (Trampoline), FreeRTOS, Zephyr
Microcontrollers: 32-bit ARM Cortex-M (STM32, TM4C), ESP32-S3, AVR (ATmega32)
Protocols & tools: UART, I2C, SPI, CAN; Git, GitHub Actions CI, CMake, Make, unit and integration testing, debugging

Education

Minia University Minia, Egypt
Bachelor of Science (B.Sc.), Computer and Systems Engineering *2020 – 2025*

- Grade: Very Good with Honors.
- Activities: Codeforces Expert; 4-time ECPC finalist (2021–2024); Treasurer, IEEE Student Branch.